



Web Development Technologies

Tutorial and Lab Sheet 5

ASP.NET Core MVC Code First - Complex Models

Exercise 1 Creating complex models

Consider the following SQL declarations:

```
create table Customer
(
    CustomerID int not null,
    Name nvarchar(50) not null,
    Address nvarchar(50) null,
    City nvarchar(40) null,
    PostCode nvarchar(4) null,
    constraint PK_Customer primary key (CustomerID)
);

create table Login
(
    LoginID char(8) not null,
    CustomerID int not null,
    PasswordHash char(94) not null,
    constraint PK_Login primary key (LoginID),
    constraint FK_Login_Customer foreign key (CustomerID) references Customer (CustomerID),
    constraint CK_Login_LoginID check (len(LoginID) = 8),
    constraint CK_Login_PasswordHash check (len>PasswordHash) = 94)
);

create table Account
(
    AccountNumber int not null,
    AccountType char not null,
    CustomerID int not null,
    Balance money not null,
    constraint PK_Account primary key (AccountNumber),
    constraint FK_Account_Customer foreign key (CustomerID) references Customer (CustomerID),
    constraint CK_Account_AccountType check (AccountType in ('C', 'S')),
    constraint CK_Account_Balance check
    (
        AccountType = 'S' and Balance >= 0 or
        AccountType = 'C' and Balance >= -100
    )
);
```



```

create table [Transaction]
(
    TransactionID int identity not null,
    TransactionType char not null,
    AccountNumber int not null,
    DestinationAccountNumber int null,
    Amount money not null,
    Comment nvarchar(30) null,
    TransactionTimeUtc datetime2 not null,
    constraint PK_Transaction primary key (TransactionID),
    constraint FK_Transaction_Account_AccountNumber
    foreign key (AccountNumber) references Account (AccountNumber),
    constraint FK_Transaction_Account_DestinationAccountNumber
    foreign key (DestinationAccountNumber) references Account (AccountNumber),
    constraint CK_Transaction_TransactionType check (TransactionType in ('D', 'W', 'T', 'S')),
    constraint CK_Transaction_Amount check (Amount > 0),
    constraint CK_Transaction_OnlyTransferCanSetDestination check
    (
        TransactionType = 'T' or
        DestinationAccountNumber is null
    ),
    constraint CK_Transaction_CannotTransferToSelf check
    (
        DestinationAccountNumber is null or
        DestinationAccountNumber != AccountNumber
    )
);

```

Your task is to do the following:

- Create a complex model in an ASP.NET Core MVC project that represents above database.
- Include some code to seed the database.
- Create Home controller with two actions: Index and Deposit.
- Here are the corresponding views:

McbaExample Home Privacy

Welcome, Matthew Bolger

Your Accounts

Account Number	Type	Balance	
4100	Saving	\$100.00	Deposit
4101	Checking	\$500.00	Deposit

McbaExample Home Privacy

Account Number 4100 (Saving)

Amount

[Deposit](#)

[Back to Accounts](#)

- The controller must include validation to ensure the amount deposited is positive and doesn't contain more than 2 decimal places, e.g., \$50.001 would not be allowed to be deposited.